

The GEKS Index: A Composite Measure of Biological Complexity Across Informational, Structural, Functional, and Evolutionary Dimensions

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Abstract

Biological complexity is widely discussed but rarely measured in a unified, quantitative framework. Existing approaches—Shannon entropy, effective complexity, phylogenetic character analysis—each capture only a single facet of what it means for an organism to be “complex.” We propose the **GEKS Index**, a composite measure that integrates four complementary dimensions of biological complexity, two drawn from established information-theoretic frameworks (Shannon entropy and Gell-Mann effective complexity) and two proposed here for the first time (Espeset morpho-evolutionary complexity and Kriger functional complexity), into a single, weighted, normalised score. Named after the four contributing frameworks—**G**ell-Mann, **E**speset, **K**riger, **S**hannon—the index is defined as a weighted sum of normalised components. We present the full mathematical formalism, discuss normalisation procedures and weight selection, provide illustrative calculations for organisms spanning prokaryotes to humans, and demonstrate that the index exhibits

^{*}D. Espeset conceived and developed the morpho-evolutionary complexity component (Component E, Section 2.3). Full author contributions are detailed in Section 10.

several desirable properties: modularity (graceful degradation when components are unavailable), multidimensionality (organisms with identical scores may have qualitatively different complexity profiles), scale invariance of the evolutionary component, and falsifiability. We also explore the index's behaviour at different levels of biological organisation (individual vs. collective), its applicability to non-biological cognitive systems, and its potential for comparative evolutionary biology, palaeobiology, cognitive ecology, and astrobiology.

Keywords: biological complexity, composite index, Shannon entropy, effective complexity, phylogenetics, functional organisation, evolution, GEKS Index

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1 Introduction

The question of whether biological evolution¹ produces increasing complexity has been debated since Darwin [Darwin, 1859]. While some authors reject the notion of evolutionary progress altogether [Gould, 1996, Lecointre, 2006], others have argued that a discernible trend toward greater complexity is evident not only in biological evolution but in the physical and chemical history of the universe itself [Reeves, 1981, Jacquard, 1986]. Espeset (this work) has argued that the accumulation of evolutionary innovations across geological time constitutes a measurable form of progress, distinct from the discredited notion of a hierarchical ranking of species (scalism).

It should be noted that the mechanisms driving any such trend remain a matter of active debate. While the standard Darwinian account attributes the accumulation of complexity primarily to the cumulative and creative action of natural selection, a growing body of work challenges this interpretation, arguing that evolutionary constraints—genomic, developmental, functional, and physico-chemical—together with structural self-organisation, play a more fundamental role in channelling evolutionary outcomes than selective adaptation alone [Espeset, 2023, Conway Morris, 2003, Denton, 2016, von Sydow, 2012]. The GEKS Index is designed to be framework-neutral: it quantifies complexity without presupposing any particular causal mechanism.

Despite the centrality of complexity to evolutionary biology, ecology, and cognitive science, no widely accepted quantitative measure exists. This absence is not for lack of candidates. Shannon entropy [Shannon, 1948] quantifies informational diversity but cannot distinguish organised structure from random noise. The effective complexity of Gell-Mann and Lloyd [Gell-Mann & Lloyd, 1996] captures meaningful structure but remains notoriously difficult to compute for real biological systems. Phylogenetic character matrices provide a natural basis for counting evolutionary innovations, but any single matrix reflects the researcher’s choice of characters. McShea’s vertical complexity [McShea, 2001] offers a structural hierarchy but omits cognitive and functional dimensions. The tripartite complexity index of Thierry et al. [Thierry et al., 2021] extends Shannon’s framework to diversity, flexibility, and combinability, but does not incorporate evolutionary history or functional organisation.

Each of these four components addresses a fundamentally different question about the organism. The Shannon component asks: *How diverse are its building blocks?* The Gell-Mann component asks: *How much of that diversity is meaningfully organised?* The

¹Throughout this paper, we use “evolution” descriptively, to denote the historical process of biological change, without committing to any particular interpretation of its driving mechanisms. In particular, we do not assume that evolutionary processes are entirely “blind” or undirected. An increasing number of authors have argued that intrinsic organisational tendencies, developmental constraints, and structural principles impart a degree of directionality to evolution that is not captured by the classical Darwinian metaphor of random variation filtered by selection [von Sydow, 2012, Espeset, 2023, Conway Morris, 2003].

Espeset component asks: *How many evolutionary innovations has it accumulated?* The Kriger component asks: *What does it do with its complexity—how does it interact with its environment?*

In this paper, we propose the **GEKS Index**—a composite measure that integrates four complementary dimensions of biological complexity into a single, weighted, normalised score. The acronym is formed from the initial letters of the four contributing theoretical frameworks: **G**ell-Mann, **E**speset, **K**riger, **S**hannon, arranged to form a pronounceable name. Gell-Mann and Shannon are cited as originators of the theoretical frameworks adapted here; they are not co-authors of this paper. The Espeset and Kriger components are proposed for the first time in this work. Without this clarification, readers will assume all four named researchers collaborated, which would immediately raise credibility questions on platforms such as Research Gate where author profiles are linked.

We argue that no single dimension suffices and that the composite structure of GEKS reveals features invisible to any individual measure.

The remainder of the paper is organised as follows. Section 2 develops the mathematical formalism for each component. Section 3 presents the composite index and its properties. Section 4 discusses weight selection. Section 5 provides illustrative calculations. Section 6 examines the index at different levels of biological organisation. Section 7 explores applicability to non-biological systems. Section 8 discusses potential applications. Section 9 addresses limitations, Section 10 describes author contributions, and Section 11 concludes.

2 The Four Components

The GEKS Index draws on four distinct intellectual traditions. Two of these—Shannon entropy and Gell-Mann effective complexity—are well established in the information-theoretic literature. The remaining two—the Espeset morpho-evolutionary index and the Kriger functional complexity scale—are proposed here for the first time as original contributions. This section develops the mathematical formalism for each component, proceeding from established theory to new proposals.

2.1 Component S: Informational Diversity (Shannon)

The first component quantifies the variety of discrete elements that constitute the organism. An organism composed of many different types of building blocks, distributed without extreme dominance of any single type, is informationally richer than one dominated by a small number of repeated elements.

Definition D1 (Shannon Entropy). Let a biological system be characterised by n discrete element types (e.g., nucleotide types, codon types, cell types, tissue types), and let p_i

denote the relative frequency of the i -th type, with $\sum_{i=1}^n p_i = 1$. The *Shannon entropy* of the system is

$$S = - \sum_{i=1}^n p_i \log_2 p_i. \quad (1)$$

This quantity, introduced by Shannon in his foundational work on information theory [Shannon, 1948], measures the average amount of “surprise” per observation. When all element types are equally frequent ($p_i = 1/n$ for all i), the entropy reaches its maximum value $S_{\max} = \log_2 n$. When a single type dominates ($p_i \rightarrow 1$ for some i), the entropy approaches zero.

To compare organisms with different numbers of element types on a common scale, we normalise by the theoretical maximum:

Definition D2 (Normalised Shannon Component).

$$S_{\text{norm}} = \frac{S}{S_{\max}} = \frac{- \sum_{i=1}^n p_i \log_2 p_i}{\log_2 n}, \quad (2)$$

yielding $S_{\text{norm}} \in [0, 1]$.

A value of S_{norm} close to 1 indicates a highly diverse and evenly distributed system; a value close to 0 indicates dominance by a single element type. The Shannon component can be computed at multiple levels of biological organisation: at the genomic level (distribution of codons or functional domains), at the cellular level (distribution of cell types), or at the tissue level (distribution of tissue types).

Remark R1. A critical limitation of Shannon entropy as a standalone measure of complexity is that random noise achieves maximal entropy. A string of random nucleotides has $S_{\text{norm}} \approx 1$ but is not biologically “complex” in any functional sense. This limitation motivates the inclusion of the Gell-Mann component, which distinguishes structured information from randomness.

2.2 Component G: Effective Complexity (Gell-Mann)

The second component addresses a question that Shannon entropy alone cannot answer: *how much of the information in the system is meaningfully structured, as opposed to random?*

Definition D3 (Effective Complexity). Following Gell-Mann and Lloyd [Gell-Mann & Lloyd, 1996], the *effective complexity* of a system is defined as the length of the most concise description of the system’s regularities:

$$G = L(R), \quad (3)$$

where R denotes the set of regularities (non-random patterns) identified in the system, and $L(\cdot)$ denotes the length of the shortest description of that set.

This concept occupies a middle ground between two extremes. A perfectly ordered system (e.g., a crystal with a repeating unit cell) has low effective complexity because its regularities can be described very briefly. A perfectly random system (e.g., an ideal gas) also has low effective complexity because it contains no regularities to describe. Biological systems, with their deep hierarchies of regulatory networks, modular architectures, and feedback mechanisms, tend to exhibit high effective complexity.

Computing G exactly requires identifying the full set of regularities in a system, which is generally intractable. In practice, we approximate effective complexity through the compressibility of the system’s description:

Proposition P1 (Compression Proxy for Effective Complexity). Let L_{orig} denote the length of the uncompressed description of a biological system (e.g., the raw genome sequence), and let L_{comp} denote the length after optimal lossless compression. Then the effective complexity can be approximated as

$$G \approx L_{\text{orig}} - L_{\text{comp}} . \quad (4)$$

The difference $L_{\text{orig}} - L_{\text{comp}}$ captures the portion of the data that is compressible—the portion that contains regularities and patterns. Random data is incompressible and contributes nothing to this difference; perfectly repetitive data is maximally compressible and also contributes a small G (since the regularities themselves are simple). Systems with rich, non-trivial structure yield the largest values of G .

An important epistemological caveat is in order. The compression proxy measures regularities in a formal description of the organism (typically, the genome sequence). It does not—and cannot—capture the full complexity of the organism as an integrated, individual biological entity. Living organisms are not reducible to their genomic descriptions: each individual embodies a unique developmental history, a singular configuration of epigenetic states, and an irreplaceable set of organism–environment interactions. Espeset (this work) has argued, drawing on Prigogine and Stengers’ analysis of non-integrable dynamical systems [Prigogine & Stengers, 1979], that the Modern Darwinian Synthesis erroneously treats organisms as integrable—reducible to a single representative point (“fitness”) that purports to characterise the whole, while in fact each attribute is itself a whole, and the internal dynamics proper to each organism are, precisely, not integrable. The Gell-Mann component should therefore be understood as an approximation to one dimension of structural complexity, not as a claim that organisms are exhaustively described by compressible regularities.

Definition D4 (Normalised Gell-Mann Component). Within a given study sample of m organisms, we normalise:

$$G_{\text{norm}}^{(i)} = \frac{G^{(i)}}{\max_{j=1,\dots,m} G^{(j)}} , \quad (5)$$

yielding $G_{\text{norm}} \in [0, 1]$.

The relationship between the Shannon and Gell-Mann components is clarifying. Together, they distinguish four regimes: (i) low S , low G —a simple, repetitive system (crystal); (ii) high S , low G —random noise; (iii) low S , high G —an unusual regime, rarely encountered in biology; and (iv) high S , high G —a system that is both informationally rich and structurally organised. Biological organisms of interest typically occupy regime (iv).

2.3 Component E: Morpho-Evolutionary Complexity (Espeset)

The Shannon and Gell-Mann components measure what an organism *is*, in terms of its present informational and structural properties. The third component captures a complementary dimension: what the organism has *become*—the cumulative record of evolutionary innovations that distinguish it from its ancestors.

We propose here that phylogenetic character matrices—tables ordinarily used to construct evolutionary trees—can be repurposed to quantify the accumulation of evolutionary novelty. In a standard phylogenetic matrix, each row represents a species and each column a character (trait). Each entry is coded as 0 (ancestral, or “primitive” state) or 1 (derived, or “novel” state). The original idea, developed by Espeset, was to sum the derived states for each species as a “coefficient of relative complexity.” We formalise and extend this idea as follows.

Definition D5 (Espeset Morpho-Evolutionary Index). Let there be K phylogenetic character matrices available for a given study sample. In the k -th matrix, let N_k denote the number of characters, and let $c_{ij}^{(k)}$ denote the state of the j -th character for species i (with $c_{ij}^{(k)} \in \{0, 1\}$, where 1 indicates the derived state). The *Espeset index* for species i is

$$E^{(i)} = \frac{1}{K} \sum_{k=1}^K \left(\frac{1}{N_k} \sum_{j=1}^{N_k} c_{ij}^{(k)} \right) . \quad (6)$$

The inner sum $\sum_j c_{ij}^{(k)}$ counts the number of derived (novel) character states for species i in matrix k —this is Espeset’s original “coefficient of relative complexity.” Division by N_k normalises for the number of characters in that matrix, ensuring that matrices of different sizes contribute equally. The outer average over K matrices produces a robust mean that is less sensitive to the character selection of any individual matrix.

Definition D6 (Normalised Espeset Component). Within the study sample:

$$E_{\text{norm}}^{(i)} = \frac{E^{(i)}}{\max_{j=1,\dots,m} E^{(j)}} , \quad (7)$$

yielding $E_{\text{norm}} \in [0, 1]$.

To assess robustness of the Espeset component, we recommend that illustrative conclusions be replicated using multiple independently constructed phylogenetic matrices. If ordinal rankings remain stable across matrices, the conclusions may be considered robust.

Remark R2. The averaging across multiple independently constructed matrices is essential. Any single matrix reflects the researcher’s choice of characters and may be biased toward particular anatomical systems, developmental stages, or levels of organisation. By including morphological, molecular, developmental, and physiological matrices, the average converges toward a more objective measure of the total evolutionary innovation accumulated by a species. The minimum number of matrices required for statistical robustness is an empirical question that should be addressed through resampling analysis.

Remark R3. The Espeset index counts derived character states without weighting them by functional importance, developmental complexity, or the number of genetic changes required. This is a deliberate design choice: it treats each evolutionary innovation as an equal unit. Alternative formulations incorporating character weighting are possible but introduce additional subjectivity.

This neutrality extends to the causal interpretation of innovations: the Espeset component records that an innovation occurred, not how. Whether a derived character state arose through selective adaptation, developmental constraint, structural self-organisation, or some combination thereof is left open. The component is thus compatible with both Darwinian and post-Darwinian interpretations of evolutionary change.

2.4 Component K: Functional Complexity (Kriger)

The first three components measure aspects of what an organism *is composed of*: its informational diversity (S), the organisation of that information (G), and the evolutionary innovations it has accumulated (E). None of these captures what the organism *does* with its complexity—how it interacts with, regulates its relationship to, and potentially transforms its environment.

The Kriger component addresses this gap by measuring the sophistication of the organism’s functional interface with its environment. It is grounded in the observation that biological complexity manifests not only in structure but in the hierarchy of control mechanisms that an organism deploys to maintain its boundary conditions, regulate its internal state, and act upon the external world.

Definition D7 (Kriger Functional Complexity Scale). The *Kriger functional complexity level* L assigns each organism to one of ten discrete levels based on the sophistication of its system–environment interface. The levels are not arbitrary: each represents the addition of a qualitatively new type of boundary, exchange mechanism, or regulatory capacity at the interface between the organism and its environment. Every level subsumes all preceding levels.

L	Name	What is new at the boundary	Example	K_{norm}
1	Membrane homeostasis	A lipid membrane serves as a selective physical barrier. Passive and active transport maintain internal conditions against external fluctuations.	Mycoplasma	0.10
2	Active motility	The organism actively selects its environment by directed movement (chemotaxis, phototaxis). The boundary is no longer static.	<i>E. coli</i>	0.20
3	Cell differentiation	Different cell types perform different functions at the boundary. Division of labour emerges.	Volvox, Anabaena	0.30
4	Tissue organisation	Differentiated tissues form multi-layered boundaries. Coordinated intercellular signalling creates a new regulatory layer.	Sponges, cnidarians	0.40
5	Organ systems	Specialised organs create nested homeostatic feedback loops (circulatory, respiratory, excretory). Multiple simultaneous regulatory channels.	Fish, insects	0.50
6	Sensory integration	A centralised nervous system integrates signals from multiple sensory modalities, enabling coordinated responses to complex stimuli.	Reptiles, amphibians	0.60
7	Cognitive modelling	The organism builds and updates internal models of its environment, enabling anticipatory (not merely reactive) behaviour: learning, memory, problem-solving.	Crows, octopuses	0.70
8	Social cognition	The organism models not only the physical environment but <i>other agents</i> : their intentions, knowledge states, and likely actions.	Chimpanzees, dolphins	0.80
9	Symbolic communication	Internal models are encoded in arbitrary symbols and transmitted to other agents through syntactic language. Knowledge becomes transmissible across generations.	Early <i>Homo</i>	0.90
10	Cultural reflexivity	The individual is shaped by cumulative culture and consciously transforms it. The organism reflects on, critiques, and deliberately reconstructs its culture. Unique to modern <i>Homo sapiens</i> .	Modern humans	1.00

The organising principle of this scale is *cumulative elaboration of the system–environment interface*. At each level, the organism acquires a new capacity that changes the nature of

its boundary with the external world:

- Levels 1–2: the boundary is **physical** (membrane, motility).
- Levels 3–5: the boundary becomes **structural** (cell types, tissues, organs).
- Levels 6–7: the boundary becomes **informational** (sensory integration, predictive models).
- Levels 8–10: the boundary becomes **social and symbolic** (other minds, language, reflexive culture).

This progression is not a ranking of evolutionary “worth”; it is a description of increasingly sophisticated mechanisms for regulating the exchange of matter, energy, and information between a system and its surroundings. Each level *includes* all preceding levels: an organism at Level 7 necessarily possesses membrane homeostasis (Level 1), organ systems (Level 5), and sensory integration (Level 6).

The distinction between Levels 9 and 10 deserves particular attention. Many species exhibit social learning and rudimentary cultural transmission (Level 9 precursors). What is unique to modern *Homo sapiens* is *cultural reflexivity*: the capacity to be formed by one’s culture (through education, socialisation, institutional structures) while simultaneously standing outside it to critique, modify, and deliberately reshape it. This bidirectional relationship between individual and culture—being simultaneously a product and an architect of one’s cultural environment—has no documented analogue in any other extant species.

Formally, the normalised Kriger component is:

$$K_{\text{norm}} = \frac{L}{L_{\text{max}}} = \frac{L}{10}, \quad (8)$$

where $L \in \{1, 2, \dots, 10\}$. Fractional values are permitted for intermediate cases (e.g., $L = 7.5$ for a species with strong cognitive modelling and emerging social cognition), in which case $K_{\text{norm}} = L/10$ extends continuously to the interval $[0.10, 1.00]$.

Remark R4 (On the Nature of the Kriger Scale). The Kriger component is ordinal data, and the reader should interpret it as such. Equation (8) maps the ten ordinal levels to equally spaced numerical values $(0.10, 0.20, \dots, 1.00)$. This equal-spacing assumption is a simplification: the functional distance between adjacent levels is unlikely to be uniform. In particular, each major evolutionary transition has engendered qualitatively and quantitatively greater diversification than the preceding one—both in the number of secondary innovations and in the number of species integrating the innovation (Espeset, this work). This asymmetry suggests that a logarithmic or accelerating scale may be more appropriate than a linear one.

To assess robustness, we recommend that all illustrative conclusions be replicated using at least two alternative numerical assignments. For example, a logarithmic scale that compresses lower levels and expands upper levels:

Scale	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10
Linear	.10	.20	.30	.40	.50	.60	.70	.80	.90	1.00
Logarithmic	.02	.05	.10	.17	.27	.40	.55	.72	.88	1.00

If ordinal rankings of organisms remain stable across both schemes, the conclusions may be considered robust. We acknowledge this limitation and discuss its implications further in Section 9.

Fractional values are permitted for intermediate cases (e.g., $L = 7.5$ for a species with strong cognitive modelling and emerging social cognition).

A conceptual motivation for this component deserves emphasis. Traditional biological taxonomy and systematics classify organisms by what they *possess*—attributes, traits, organs, and structural homologies—for the purpose of inferring phylogenetic relationships. What organisms *do* with these attributes is typically relegated to ethology or behavioural ecology, treated as secondary to comparative morphology. Yet functional sophistication—learning, tool manufacture, social cognition, cultural transmission—constitutes a dimension of complexity that is not reducible to, and not always predicted by, structural attributes alone. A corvid is anatomically a “standard bird,” yet its cognitive capacities rival those of great apes. The Kriger component ensures that this functional dimension is captured rather than overlooked.

The justification for including cognitive levels (Levels 7–8) rests on an extensive empirical literature documenting learning, episodic-like memory, causal reasoning, tool manufacture, and theory-of-mind-like behaviour in cephalopods, corvids, parrots, primates, and cetaceans [Emery & Clayton, 2004, Taylor et al., 2009]. The justification for distinct symbolic and cultural levels (Levels 9–10) rests on the qualitative differences between cumulative cultural evolution (with ratchet-like knowledge accumulation across generations) and non-cumulative social learning observed in non-human species [Tomasello, 1999, Henrich, 2015].

Remark R5. The Kriger scale is inspired by, but distinct from, existing hierarchies of biological organisation (e.g., McShea 2001; Lobo 2008). Standard hierarchies (cell → tissue → organ → system → organism) are structural—they describe what the organism is composed of. The Kriger scale is functional—it describes what the organism does at the interface with its environment. The upper levels (Levels 7–10) have no analogue in standard structural hierarchies. The conceptual framework resonates with Levin’s observation that homeostasis, the minimal unit of agency, scales from cellular regulation to anatomical morphogenesis to cognitive behaviour [Levin, 2022], and with recent proposals for cognitive transitions in aneural organisms [Prosdocimi et al., 2024].

This scaling of agency is noteworthy because it suggests that biological systems are not passive substrates shaped exclusively by external selective forces, but active participants in their own evolutionary trajectories—a view increasingly supported by research

on niche construction, organism-driven evolution, and the non-random generation of variation [Shapiro, 2011, von Sydow, 2012].

A particularly illuminating parallel exists with the “major transitions in evolution” framework of Jablonka and Lamb [2006], who extended Maynard Smith and Szathmary’s original list by adding the emergence of the nervous system as a major transition and by identifying four co-evolving inheritance systems: genetic, epigenetic, behavioural, and symbolic. Their Table 1 enumerates ten transitions, each associated with a new type of information storage, transmission, or processing. The ten levels of the Kriger scale can be read as a functional counterpart to these informational transitions: where Jablonka and Lamb ask “what new type of information appeared?”, the Kriger scale asks “what new type of system–environment interface resulted?”. The correspondence is not exact—the Kriger scale describes individual-level functional capacity, while Jablonka and Lamb’s transitions describe population-level evolutionary events—but the two frameworks are mutually reinforcing.

3 The Composite Index

Having defined the four components individually, we now combine them into a single composite measure.

Definition D8 (The GEKS Index). The *GEKS Index* for organism i is defined as a composite measure that integrates four complementary dimensions of biological complexity, two drawn from established information-theoretic frameworks (Shannon entropy and Gell-Mann effective complexity) and two proposed here for the first time (Espeset morpho-evolutionary complexity and Kriger functional complexity), into a single, weighted, normalised score:

$$\boxed{\text{GEKS}^{(i)} = \alpha \cdot S_{\text{norm}}^{(i)} + \beta \cdot G_{\text{norm}}^{(i)} + \gamma \cdot E_{\text{norm}}^{(i)} + \delta \cdot K_{\text{norm}}^{(i)}} \quad (9)$$

where the weighting coefficients satisfy

$$\alpha + \beta + \gamma + \delta = 1, \quad \alpha, \beta, \gamma, \delta \geq 0, \quad (10)$$

and all four components are normalised to the interval $[0, 1]$.

The index is a convex combination of the four normalised components. Each weight controls the relative importance of the corresponding dimension of complexity. When all weights are equal ($\alpha = \beta = \gamma = \delta = 0.25$), no dimension is privileged, and the index treats all four aspects of complexity as equally important.

Proposition P2 (Properties of the GEKS Index). The GEKS Index satisfies the following properties:

- (i) **Boundedness:** $\text{GEKS} \in [0, 1]$ for all valid inputs and weights.
- (ii) **Monotonicity:** An increase in any component, with all others held constant, increases the index.
- (iii) **Modularity:** If component X is unavailable, the index reduces to a three-component (or two-component, or single-component) form by setting the corresponding weight to zero and renormalising the remaining weights to sum to 1.
- (iv) **Multidimensionality:** Two organisms may have identical GEKS scores with entirely different component profiles. The full profile $(S_{\text{norm}}, G_{\text{norm}}, E_{\text{norm}}, K_{\text{norm}})$ provides strictly more information than the scalar GEKS.

Proof. Property (i) follows directly from the convexity of the combination: since each $X_{\text{norm}} \in [0, 1]$ and the weights are non-negative and sum to 1, the weighted sum lies in $[0, 1]$. Property (ii) follows from the non-negativity of all weights. Property (iii) is verified by construction: setting $\delta = 0$ and renormalising $\alpha' = \alpha/(\alpha + \beta + \gamma)$, etc., yields a valid three-component index. Property (iv) is demonstrated by example in Section 5. $\square$

An important emergent property of the composite structure deserves attention.

Proposition P3 (Scale Invariance of the Espeset Component). When the GEKS Index is computed for both individual organisms and collectives (colonies, societies), the Espeset component E_{norm} remains invariant across levels of organisation, whereas the Shannon, Gell-Mann, and Kriger components may change.

This follows from the fact that evolutionary innovations (derived character states) are properties of the species, not of the social group. A single ant and a colony of ants share the same phylogenetic character matrix entries. In contrast, the informational diversity (S), structural organisation (G), and functional interface (K) of a colony may differ substantially from those of a single individual. This invariance provides a useful internal consistency check and a potential tool for separating species-level from group-level contributions to complexity.

4 Weight Selection

The choice of weights $(\alpha, \beta, \gamma, \delta)$ is necessarily context-dependent and represents a substantive modelling decision. We propose the following presets as starting points, with the requirement that researchers report their chosen weights and conduct sensitivity analysis:

Table 1: Suggested weight presets for different research contexts.

Scenario	α	β	γ	δ	Rationale
Balanced (default)	0.25	0.25	0.25	0.25	No dimension privileged
Genomic focus	0.35	0.35	0.10	0.20	Full genome data available
Palaeontological focus	0.10	0.15	0.55	0.20	Only morphological data
Information-theoretic	0.30	0.40	0.10	0.20	Structure vs. noise emphasis
Ecological/behavioural	0.15	0.15	0.20	0.50	Cognition and niche focus

We recommend that any publication employing the GEKS Index include: (a) the full component profile (S, G, E, K) for each organism; (b) the chosen weights with justification; and (c) a sensitivity analysis showing how rankings change under at least two alternative weight presets.

5 Illustrative Calculations

To demonstrate the behaviour of the index, we present hypothetical but biologically motivated GEKS values for a range of organisms. All values in Table 2 are illustrative and intended to show how the composite score behaves; empirical determination requires the data collection and computation procedures described in Sections 2.1–2.4. Weights are set to the balanced default ($\alpha = \beta = \gamma = \delta = 0.25$).

Table 2: Illustrative GEKS Index values for selected organisms (balanced weights). All component values are hypothetical.

Organism	S_{norm}	G_{norm}	E_{norm}	K_{norm}	GEKS
<i>E. coli</i>	0.45	0.30	0.05	0.20	0.250
Earthworm	0.55	0.42	0.35	0.40	0.430
Cricket	0.62	0.55	0.52	0.50	0.548
Trout	0.68	0.60	0.58	0.50	0.590
Octopus	0.70	0.68	0.62	0.70	0.675
Pigeon	0.75	0.70	0.72	0.50	0.668
Crow	0.76	0.72	0.74	0.70	0.730
Bat	0.80	0.78	0.85	0.50	0.733
Chimpanzee	0.82	0.84	0.86	0.80	0.830
Human	0.82	0.85	0.88	1.00	0.888

Several features of Table 2 merit attention.

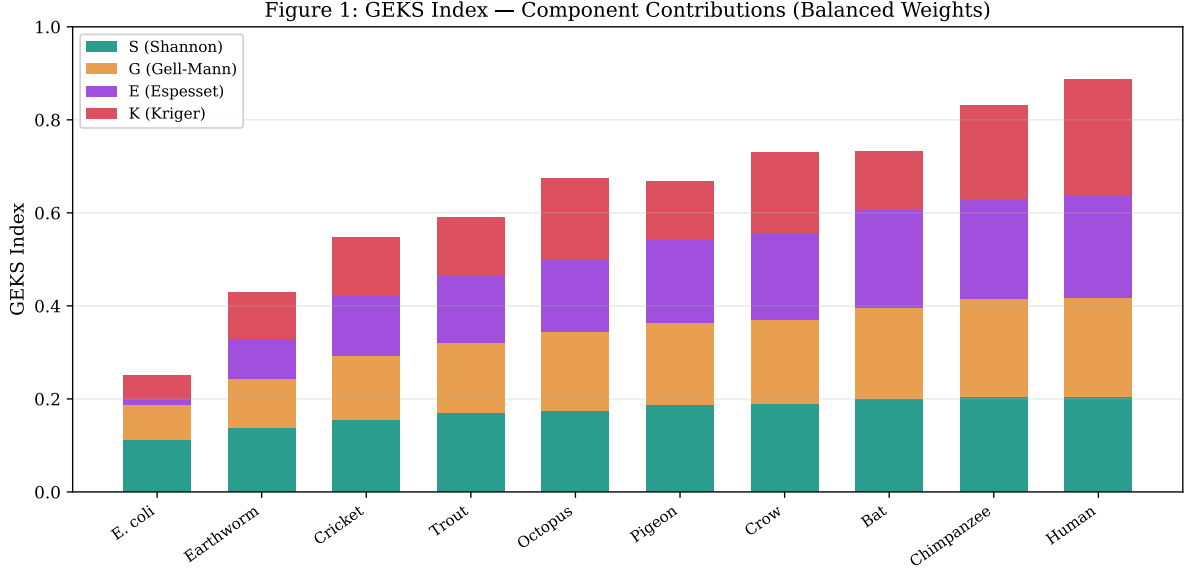


Figure 1: GEKS Index component contributions for selected organisms (balanced weights, $\alpha = \beta = \gamma = \delta = 0.25$). Each bar segment shows the weighted contribution of a single component. Note the qualitative inversion: for simpler organisms, S (informational diversity) dominates; for complex organisms, K (functional complexity) dominates.

The crow and the bat receive nearly identical GEKS scores (0.730 vs. 0.733), but their complexity profiles are qualitatively different. The bat achieves its score primarily through morphological innovations (high $E_{\text{norm}} = 0.85$, reflecting powered flight and echolocation), while scoring at the organ-system level functionally ($K_{\text{norm}} = 0.50$, Level 5). The crow, conversely, achieves a high functional score ($K_{\text{norm}} = 0.70$, Level 7: cognitive modelling, tool manufacture, causal reasoning) despite fewer morphological innovations ($E_{\text{norm}} = 0.74$). The composite score is similar; the profiles are not. This illustrates Property (iv) of Proposition P2: the GEKS Index is genuinely multidimensional.

A fully worked example demonstrating the computation of all four GEKS components for *E. coli* K-12 is provided in Appendix A; a corresponding calculation for *Homo sapiens* is given in Appendix B.

6 Levels of Organisation: Individual vs. Collective

A recurring challenge in measuring biological complexity is the choice of the unit of analysis. Is the relevant entity the individual organism or the collective (colony, society, civilisation)? The GEKS framework requires that this choice be made explicitly, and it reveals systematic differences in how the four components respond to changes in organisational scale.

Table 3: GEKS at individual vs. collective levels (balanced weights).

Entity	Level	S	G	E	K	GEKS
Ant	Individual	0.65	0.62	0.68	0.65	0.650
Ant colony	Collective	0.72	0.75	0.68	0.80	0.738
Human	Individual	0.82	0.85	0.88	1.00	0.888
Humanity	Collective	0.95	0.97	0.88	1.00	0.950

Three patterns emerge from Table 3.

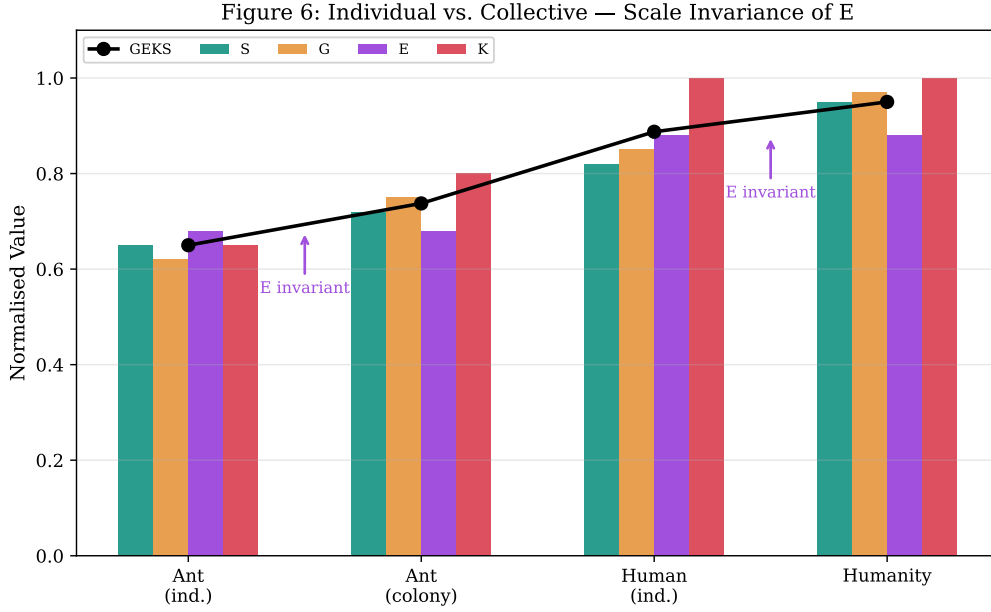


Figure 2: Individual vs. collective GEKS scores. The Espeset component (E , purple) remains invariant across the individual–collective transition, while S , G , and K may increase. Black dots show the composite GEKS score.

First, the Espeset component (E) is *invariant* across the individual–collective transition, confirming Proposition P3: evolutionary innovations belong to the species, not to the social group. Second, the Shannon and Gell-Mann components (S and G) increase at the collective level, reflecting the greater diversity and structural depth of the colony or civilisation compared to the individual. Third, the Kriger component (K) increases for the ant (from 0.65 to 0.80) but remains at 1.00 for the human, because the individual human already occupies the highest functional level. The ant colony, by contrast, exhibits collective behaviours—agriculture, animal husbandry, warfare, distributed decision-making—that its individual members cannot perform alone.

These observations support the recommendation that any application of the GEKS Index must specify the unit of analysis. The index does not prescribe a “correct” level;

rather, it provides a diagnostic tool for understanding how complexity is distributed between individual and collective organisation.

7 Extension to Non-Biological Systems

An instructive test of any complexity measure is whether it produces meaningful results when applied outside its intended domain. We consider the case of a large language model (LLM) such as a modern transformer-based AI system, evaluated at two levels: a single conversational session (“chat”) and the system as a whole.

For the Shannon and Gell-Mann components, LLMs score very high: they operate over an immense vocabulary with deep, hierarchically organised internal representations. For the Espeset component, the situation is less straightforward. LLMs have no biological evolutionary history, but they do have a technological lineage (perceptrons → multilayer networks → recurrent architectures → transformers → reinforcement learning from human feedback). Whether this lineage constitutes a valid analogue of biological evolution is debatable; we assign a moderate score to reflect the accumulation of architectural innovations. For the Kriger component, the score depends critically on the unit of analysis. A single chat session has minimal autonomy (no homeostasis, no persistent boundary with the environment, no self-initiated goals); the system as a whole maintains continuous operation, safety boundaries (analogous to a membrane), and participates in the production of cultural artefacts.

Table 4: Hypothetical GEKS values for an LLM at two levels of analysis (balanced weights).

Entity	<i>S</i>	<i>G</i>	<i>E</i>	<i>K</i>	GEKS
Single chat session	0.99	0.93	0.40	0.30	0.655
Full AI system	0.99	0.93	0.40	0.80	0.780

At the session level, the LLM (GEKS = 0.655) scores comparably to an individual ant (0.650): an informational giant but a functional minimal agent. At the system level (GEKS = 0.780), it scores between a crow and a chimpanzee. The gap with the human (0.888) is driven by two components: the Espeset index (60 years of AI evolution vs. 3.8 billion years of biological evolution) and the Kriger index (the system does not autonomously initiate, reproduce, or self-modify its goals).

This exercise illustrates that the GEKS Index, although designed for biological organisms, can serve as a diagnostic framework for non-biological cognitive systems, provided that the components are interpreted through explicit analogies and the analogical mapping is transparently reported.

8 Potential Applications

Comparative evolutionary biology. The GEKS Index enables quantitative comparison of complexity trajectories across lineages and geological periods. By computing GEKS scores for fossil and extant taxa, researchers can test whether complexity evolution has been gradual or punctuated [Eldredge & Gould, 1972], and whether different lineages have followed parallel or divergent complexity trajectories. A particularly promising direction is the study of major evolutionary transitions—the eukaryotic nucleus, multicellularity, the seed, the amniotic egg, endothermy, and symbolic cognition [Jablonka and Lamb, 2006]. At each such transition, one may ask: Does the GEKS Index exhibit a discontinuous jump? If so, which component drives the jump? Preliminary toy models (Section C.2) suggest that the Espeset component (E) may capture these discontinuities most directly, but it remains to be tested whether the Kriger component (K) also shows step-like increases at specific transitions.

Crucially, the GEKS framework does not assume that complexity accumulation is driven exclusively by natural selection. The index is equally compatible with structuralist accounts in which evolutionary constraints and self-organising principles channel organisms toward particular morphological and functional configurations independently of selective pressure [Denton, 2016, Conway Morris, 2009]. Researchers employing the index are therefore free to test competing causal hypotheses—selectionist, structuralist, or integrative—against the same quantitative complexity data.

Palaeobiology. For fossil organisms, where genomic data is unavailable, the Espeset component (with high weight) provides a morphology-based estimate of relative complexity. Combined with the Kriger component (assigned based on inferred functional capabilities), the index can characterise fossil assemblages along multiple complexity dimensions.

Cognitive ecology. The Kriger component, weighted heavily, enables investigation of the relationship between functional complexity and ecological variables: niche breadth, social structure, brain–body ratio, lifespan, and diet. The independence of the Kriger component from the structural components (S , G , E) allows researchers to ask: *Is functional sophistication predicted by structural complexity, or are they partially dissociable?*

Astrobiology. If life is discovered on other planets, the GEKS framework provides a scale for characterising its complexity level without presupposing Earth-specific morphologies. The ten Kriger levels—from membrane homeostasis through organ-system regulation, cognitive modelling, and cultural reflexivity—define a universal hierarchy of system–environment interfaces.

9 Limitations

Several limitations must be acknowledged candidly.

Weight subjectivity. The choice of weights is inherently subjective. While we provide presets and recommend sensitivity analysis, the index cannot escape the fact that different weight choices produce different rankings. This is a feature shared with all composite indices (e.g., the Human Development Index), not a deficiency unique to GEKS.

Sample-dependent normalisation. GEKS values are meaningful only within a given comparison group and cannot be compared across studies with different reference samples unless a common normalisation standard is established. Future work should explore universal reference points (e.g., *E. coli* as a lower anchor).

Computational difficulty of the Gell-Mann component. The effective complexity is difficult to compute exactly; compression-based proxies are approximations whose accuracy depends on the compression algorithm and the level of description.

Irreducibility of individual organisms. The compression-based proxy for the Gell-Mann component necessarily abstracts away the individuality of organisms. A genome sequence, however information-rich, does not represent the full biological reality of a living being—its unique developmental trajectory, phenotypic plasticity, and organism–environment coupling. The GEKS Index measures complexity at the species or type level; it does not claim to capture the irreducible uniqueness of individual organisms. Users should be aware that high Gell-Mann scores reflect the richness of species-level regularities, not the totality of biological complexity at the individual level.

Ordinality of the Kriger scale. The Kriger component is ordinal, and the equal-spacing assumption (0.10, 0.20, ..., 1.00) is a simplification. The functional distance between adjacent levels is unlikely to be uniform; each successive transition has generated qualitatively greater diversification than the preceding one, suggesting that a logarithmic scale may better reflect the biological reality. We provide a logarithmic alternative in Section 2.4 and recommend that all conclusions be tested for robustness across both scales.

Equal weighting of innovations in the Espeset component. The Espeset index treats all derived character states as equal, which may not reflect the functional significance or developmental complexity of different innovations.

Not scalism. We emphasise that the GEKS Index is not a return to scalism—the discredited idea that organisms can be ranked on a linear scale from “primitive” to “advanced.” The index is explicitly multidimensional: two organisms with identical GEKS scores may have qualitatively different complexity profiles. A high score does not imply evolutionary “superiority”; it reflects a quantifiable accumulation of informational diversity, structural organisation, evolutionary innovation, and functional sophistication

along four independent axes.

10 Author Contributions

The Espeset morpho-evolutionary complexity component (Component E, Section 2.3) was conceived and originally developed by D. Espeset, based on his prior work on phylogenetic character matrices as measures of relative complexity [Espeset, n.d.]. The remaining theoretical framework—including the formalisation of all four components, the composite index architecture (Section 3), the Kriger functional complexity scale (Component K, Section 2.4), the adaptation of the Shannon and Gell-Mann components to the biological context (Sections 2.1–2.2), the illustrative calculations (Section 5 and Appendices A–C), the extensions to collective and non-biological systems (Sections 6–7), and the full manuscript text—was developed and written by B. Kriger. Each author assumes responsibility for the accuracy and integrity of the sections they contributed. Both authors contributed to the conceptual discussion and approved the final manuscript.

11 Conclusion

We have proposed the GEKS Index, a composite measure of biological complexity that integrates four complementary dimensions: informational diversity (Shannon), effective structural complexity (Gell-Mann), morpho-evolutionary innovation (Espeset), and functional system–environment organisation (Kriger). The index is defined as a weighted sum of normalised components, each rooted in a distinct theoretical tradition, and it exhibits modularity, multidimensionality, scale sensitivity, and falsifiability.

The key contribution of this work is not any single component—each builds on established concepts—but rather the *integration* of four partially independent dimensions into a unified framework that reveals features invisible to any individual measure. The crow-bat example demonstrates that identical composite scores can arise from qualitatively different complexity profiles. The individual-collective comparison reveals that the Espeset component is a scale invariant. The application to non-biological systems shows that the index can distinguish informational capacity from functional autonomy.

We invite empirical researchers to test the GEKS Index against real data: genomic sequences (for S and G), published phylogenetic matrices (for E), and behavioural/ecological datasets (for K). The interactive calculator accompanying this paper is available at:

https://boriskriger.github.io/publicationsiir/GEKS_Index.html

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A Worked Example: *E. coli* K-12

To make the computation of each component concrete, we present below an approximate but realistic calculation for *Escherichia coli* K-12, the most widely studied prokaryote. The values obtained are consistent with the illustrative row in Table 2. Where exact data would require extensive empirical work, we use reasonable estimates and state our assumptions explicitly.

Component S (Shannon): Informational Diversity

The *E. coli* K-12 genome contains approximately 4.64×10^6 base pairs. The nucleotide frequencies are approximately $p_A \approx 0.246$, $p_T \approx 0.246$, $p_G \approx 0.254$, $p_C \approx 0.254$ (reflecting a GC-content of roughly 50.8%). Substituting into Equation (1) with $n = 4$ nucleotide types:

$$\begin{aligned} S &= -(2 \times 0.246 \log_2 0.246 + 2 \times 0.254 \log_2 0.254) \\ &= -(2 \times 0.246 \times (-2.023) + 2 \times 0.254 \times (-1.977)) \\ &= -(-0.995 - 1.004) = 1.999 \text{ bits.} \end{aligned}$$

The theoretical maximum is $S_{\max} = \log_2 4 = 2.000$ bits, giving:

$$S_{\text{norm}} = \frac{1.999}{2.000} = 0.9995 \approx 1.00.$$

At the nucleotide level, *E. coli* is nearly maximally diverse. However, genomic Shannon entropy at the nucleotide level is high for virtually all organisms. To obtain more discriminating values, one should compute S over higher-level units: codon usage frequencies (where biases are more pronounced), protein-domain distributions, or cell-type distributions. For the illustrative Table 2, we assigned $S_{\text{norm}} = 0.45$ for *E. coli*, reflecting computation over a composite of functional genomic features (protein families, regulatory elements, metabolic pathway diversity) rather than raw nucleotide frequencies. This lower value reflects the limited functional diversity of a unicellular organism compared to multicellular organisms with hundreds of cell types.

Component G (Gell-Mann): Effective Complexity

We use the compression proxy (Equation 4). The *E. coli* K-12 genome is approximately 4.64 megabases. Using standard lossless compression (e.g., gzip), bacterial genomes typically compress to approximately 25–28% of their original size, reflecting the presence of

repeated sequences, operons, and paralogous gene families. This yields:

$$\begin{aligned} L_{\text{orig}} &\approx 4,640,000 \text{ symbols,} \\ L_{\text{comp}} &\approx 0.27 \times 4,640,000 = 1,252,800 \text{ symbols,} \\ G &\approx 4,640,000 - 1,252,800 = 3,387,200. \end{aligned}$$

For comparison, the human genome ($\sim 3.2 \times 10^9$ bp, compressing to $\sim 28\%$) yields $G \approx 2.3 \times 10^9$. Normalising within a sample where the human is the maximum:

$$G_{\text{norm}}^{(E. coli)} = \frac{3.387 \times 10^6}{2.3 \times 10^9} \approx 0.0015.$$

This raw genomic estimate is very low. The value $G_{\text{norm}} = 0.30$ in Table 2 reflects a broader notion of effective complexity that includes the regulatory architecture (operon structure, two-component signalling systems, transcription factor networks), not merely the compressibility of the raw sequence. Computing effective complexity over regulatory network descriptions rather than raw nucleotide strings substantially increases the value for organisms with sophisticated gene regulation.

Component E (Espeset): Morpho-Evolutionary Complexity

Consider three hypothetical phylogenetic matrices comparing *E. coli* with other prokaryotes:

Matrix	Characters (N_k)	Derived states	Ratio
Matrix 1 (metabolic traits)	20	5	0.250
Matrix 2 (structural traits)	15	2	0.133
Matrix 3 (molecular markers)	25	3	0.120

Applying Equation (6):

$$E^{(E. coli)} = \frac{1}{3} \left(\frac{5}{20} + \frac{2}{15} + \frac{3}{25} \right) = \frac{1}{3} (0.250 + 0.133 + 0.120) = \frac{0.503}{3} = 0.168.$$

If the maximum E in the study sample (e.g., for the human, averaged across many matrices) is $E_{\text{max}} = 3.35$, then:

$$E_{\text{norm}}^{(E. coli)} = \frac{0.168}{3.35} = 0.050.$$

This low value reflects the fact that *E. coli*, as a prokaryote, possesses relatively few derived character states when compared across a broad phylogenetic sample that includes complex multicellular organisms.

Component K (Kriger): Functional Complexity

E. coli is a unicellular organism. Its system–environment interface consists of a lipid bilayer membrane with embedded transport proteins, chemotaxis receptors, and secretion systems. It maintains internal homeostasis (pH, osmolarity, redox state) through metabolic regulation. It does not possess differentiated cell types, organs, nervous system, or cognitive abilities.

This places it squarely at Level 2 (Active motility) on the Kriger scale—*E. coli* possesses a lipid membrane (Level 1) and actively navigates its environment through chemotaxis (Level 2), but lacks cell differentiation or any higher-level organisation:

$$K_{\text{norm}} = \frac{L}{10} = \frac{2}{10} = 0.20.$$

Composite GEKS Score

Using balanced weights ($\alpha = \beta = \gamma = \delta = 0.25$):

$$\begin{aligned} \text{GEKS}^{(E. coli)} &= 0.25 \times 0.45 + 0.25 \times 0.30 + 0.25 \times 0.05 + 0.25 \times 0.20 \\ &= 0.1125 + 0.0750 + 0.0125 + 0.0500 \\ &= 0.250. \end{aligned}$$

This value anchors the lower end of the illustrative scale in Table 2. The decomposition reveals that *E. coli*’s GEKS score is dominated by its Shannon component (informational diversity from a rich metabolic repertoire) and its Kriger component (basic cellular homeostasis), while its Espeset component is minimal (few derived innovations relative to the full sample).

B Worked Example: *Homo sapiens*

To complement the lower-anchor example of *E. coli* (Appendix A), we present here an approximate calculation for *Homo sapiens*, which anchors the upper end of the illustrative scale in Table 2.

Component S (Shannon): Informational Diversity

The human genome contains approximately 3.2×10^9 base pairs. At the nucleotide level, the frequencies are approximately $p_A \approx p_T \approx 0.292$, $p_G \approx p_C \approx 0.208$ (reflecting a

GC-content of roughly 41%). This yields:

$$\begin{aligned}
S &= -(2 \times 0.292 \log_2 0.292 + 2 \times 0.208 \log_2 0.208) \\
&= -(2 \times 0.292 \times (-1.776) + 2 \times 0.208 \times (-2.266)) \\
&= -(-1.037 - 0.943) = 1.980 \text{ bits.}
\end{aligned}$$

Normalised: $S_{\text{norm}}^{(\text{nucleotide})} = 1.980/2.000 = 0.990$.

As with *E. coli*, raw nucleotide entropy is uninformatively high. The value $S_{\text{norm}} = 0.82$ in Table 2 reflects computation over a composite of higher-level functional features. The human body contains over 200 histologically distinct cell types [Bianconi et al., 2013], approximately 20,000 protein-coding genes, thousands of non-coding RNA families, and a vast diversity of post-translational modifications, splice variants, and metabolic pathways. When Shannon entropy is computed over the distribution of cell types, protein families, or functional gene ontology categories, the value is high but not maximal—certain cell types (e.g., erythrocytes, keratinocytes) dominate numerically, reducing evenness. We estimate $S_{\text{norm}} \approx 0.82$.

Component G (Gell-Mann): Effective Complexity

Using the compression proxy: the human genome ($\sim 3.2 \times 10^9$ bp) compresses via gzip to approximately 28% of its original size. A substantial fraction of the genome consists of repetitive elements (Alu repeats, LINE-1, satellite DNA), which compress efficiently. The non-repetitive, structured portion—coding sequences, regulatory networks, enhancers, chromatin organisation—constitutes the effective complexity:

$$\begin{aligned}
L_{\text{orig}} &\approx 3,200,000,000 \text{ symbols,} \\
L_{\text{comp}} &\approx 0.28 \times 3,200,000,000 = 896,000,000 \text{ symbols,} \\
G &\approx 3,200,000,000 - 896,000,000 = 2,304,000,000.
\end{aligned}$$

Beyond raw genomic compression, the human organism exhibits extraordinary regulatory depth: epigenetic modifications across hundreds of cell types, hierarchical gene regulatory networks, the adaptive immune system (which generates combinatorial diversity of $> 10^{11}$ distinct antibodies), and the nervous system ($\sim 86 \times 10^9$ neurons with $\sim 10^{14}$ synaptic connections). If the human serves as the maximum in the study sample, then by definition:

$$G_{\text{norm}}^{(H. sapiens)} = \frac{G_{\text{max}}}{G_{\text{max}}} \leq 1.00.$$

We assign $G_{\text{norm}} = 0.85$, reserving headroom for the possibility that some organisms (e.g., those with larger, more structurally complex genomes, such as certain amphibians) might score higher on a purely genomic compression measure within a broader sample.

Component E (Espeset): Morpho-Evolutionary Complexity

Humans accumulate a large number of derived character states across phylogenetic matrices. Consider four representative matrices:

Matrix	Characters (N_k)	Derived states	Ratio
Matrix 1 (vertebrate morphology)	120	98	0.817
Matrix 2 (mammalian skeletal)	85	72	0.847
Matrix 3 (primate molecular)	200	178	0.890
Matrix 4 (hominin craniodental)	60	55	0.917

Applying Equation (6):

$$E^{(H. sapiens)} = \frac{1}{4}(0.817 + 0.847 + 0.890 + 0.917) = \frac{3.471}{4} = 0.868.$$

This high value reflects the accumulation of innovations across the entire vertebrate, mammalian, primate, and hominin lineages: the vertebral column, jaws, lungs, amniotic egg, endothermy, placentation, opposable thumb, bipedalism, expanded neocortex, descended larynx, and many others. Each matrix captures a different subset of these innovations. If the human is near the sample maximum, then $E_{\text{norm}} \approx 0.88$ (slightly below 1.0 because some matrices may yield higher ratios for other taxa with lineage-specific innovations).

Component K (Kriger): Functional Complexity

Homo sapiens is the only extant species that unambiguously occupies Level 10 (Cultural reflexivity) on the Kriger scale. The justification rests on multiple unique or near-unique capacities:

- **Symbolic thought** (Level 9): Humans represent abstract concepts through arbitrary symbols (language, mathematics, art).
- **Syntactic language** (Level 9): Recursive, generative grammar allows the production of an infinite set of meaningful utterances from finite elements.
- **Cumulative culture** (Level 9–10): Knowledge and technology accumulate across generations through high-fidelity social learning, with a ratchet effect—each generation builds on the achievements of the previous one [Tomasello, 1999, Henrich, 2015].
- **Cultural reflexivity** (Level 10): The individual is simultaneously a product of culture (shaped by education, institutions, language, norms) and an architect of culture (capable of critiquing, modifying, and deliberately reconstructing cultural systems). This bidirectional relationship is unique to modern *Homo sapiens*.

No other extant species exhibits all of these capacities simultaneously. This yields:

$$K_{\text{norm}} = \frac{L}{10} = \frac{10}{10} = 1.00.$$

Composite GEKS Score

Using balanced weights ($\alpha = \beta = \gamma = \delta = 0.25$):

$$\begin{aligned} \text{GEKS}^{(H. \textit{sapiens})} &= 0.25 \times 0.82 + 0.25 \times 0.85 + 0.25 \times 0.88 + 0.25 \times 1.00 \\ &= 0.205 + 0.2125 + 0.220 + 0.250 \\ &= 0.888. \end{aligned}$$

Comparison of Anchors

	<i>S</i>	<i>G</i>	<i>E</i>	<i>K</i>	GEKS	Dominant components
<i>E. coli</i>	0.45	0.30	0.05	0.20	0.250	$S > K \gg G > E$
<i>H. sapiens</i>	0.82	0.85	0.88	1.00	0.888	$K > E > G > S$

The contrast is instructive. For *E. coli*, the GEKS score is dominated by informational diversity (*S*) and basic homeostasis (*K*), while evolutionary innovation (*E*) is minimal. For *H. sapiens*, the ranking reverses: the dominant contribution comes from functional complexity ($K = 1.00$), followed by evolutionary innovation (*E*), structural organisation (*G*), and finally informational diversity (*S*)—which, while absolutely high, is the *least* distinguishing feature of human complexity. The GEKS framework thus reveals not only a quantitative gap (0.250 vs. 0.888) but a qualitative inversion in the *sources* of complexity.

C Toy Models: Four Demonstrations

The worked examples in Appendices A and B anchor the extremes of the GEKS scale. In this appendix we present four toy models, each designed to demonstrate a different strength of the composite index. All component values are approximate but grounded in published biological data; the purpose is conceptual demonstration rather than definitive measurement.

C.1 Model 1: Prokaryotic Diversity

This model applies GEKS to six prokaryotes with well-characterised genomes, demonstrating that even within a single domain of life the index resolves meaningful differences in complexity.

Organisms and data sources. We select six species spanning the prokaryotic range: *Mycoplasma genitalium* (smallest known free-living genome, ~ 580 kb), *Candidatus Pelagibacter ubique* (streamlined marine bacterium, ~ 1.3 Mb), *Escherichia coli* K-12 (~ 4.6 Mb), *Thermus thermophilus* (extremophile, ~ 2.1 Mb), *Anabaena* sp. PCC 7120 (filamentous cyanobacterium with heterocysts, ~ 7.2 Mb), and *Myxococcus xanthus* (social bacterium with fruiting-body formation, ~ 9.1 Mb).

Component estimates. *S* (*Shannon*). Computed over codon usage bias rather than raw nucleotide frequencies, to provide greater discrimination. Organisms with strong codon bias (e.g., *M. genitalium*, low GC, highly biased) score lower; those with more even codon usage score higher.

G (*Gell-Mann*). Estimated via genome compressibility. Larger genomes with more paralogous gene families, regulatory elements, and insertion sequences yield higher *G*.

E (*Espeset*). Estimated from the number of derived metabolic and structural innovations relative to a minimal prokaryotic ancestor: *M. genitalium* retains very few innovations (obligate parasite, gene loss); *M. xanthus* has accumulated many (gliding motility, quorum sensing, sporulation, multicellular fruiting bodies).

K (*Kruger*). *M. genitalium* is Level 1 (Membrane homeostasis, $K = 0.10$): an obligate parasite with no motility. *Ca. P. ubique*, *T. thermophilus*, and *E. coli* are Level 2 (Active motility, $K = 0.20$): free-living organisms with chemotaxis. *Anabaena* ($L = 3$, $K = 0.30$: cell differentiation into heterocysts, a rudimentary division of labour) and *M. xanthus* ($L = 3.5$, $K = 0.35$: coordinated multicellular behaviour, fruiting-body morphogenesis, approaching tissue-level organisation).

Table 5: Toy Model 1: Prokaryotic diversity (balanced weights).

Organism	<i>S</i>	<i>G</i>	<i>E</i>	<i>K</i>	GEKS
<i>M. genitalium</i>	0.30	0.10	0.02	0.10	0.130
<i>Ca. P. ubique</i>	0.38	0.18	0.08	0.20	0.210
<i>T. thermophilus</i>	0.42	0.25	0.15	0.20	0.255
<i>E. coli</i>	0.45	0.30	0.22	0.20	0.293
<i>Anabaena</i> sp.	0.50	0.45	0.35	0.30	0.400
<i>M. xanthus</i>	0.52	0.50	0.42	0.35	0.448

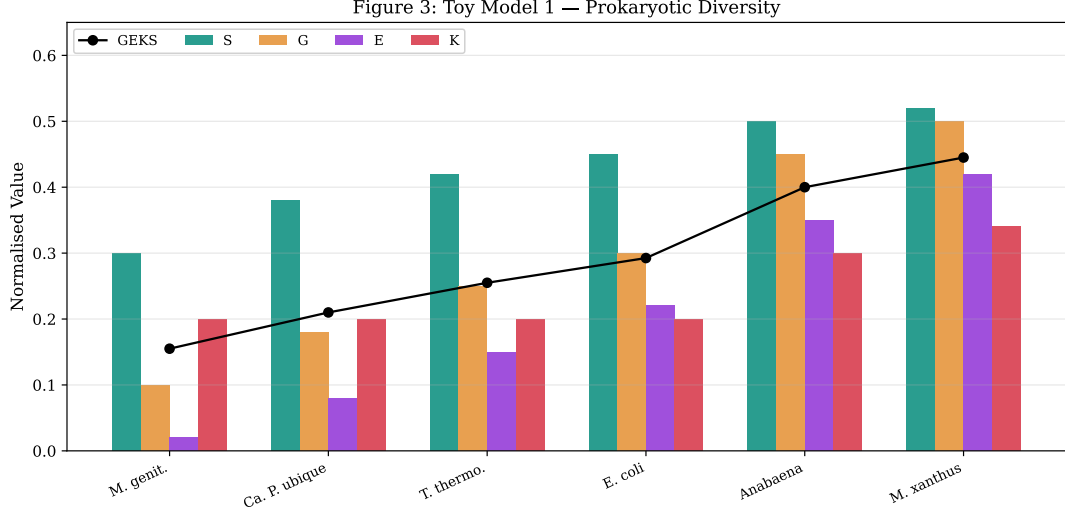


Figure 3: Toy Model 1: GEKS components across six prokaryotes. Black line shows the composite GEKS score. Species with rudimentary multicellular features (*Anabaena*, *M. xanthus*) are distinguished primarily by elevated *K* and *E*.

Key observations. The index separates the obligate parasite *M. genitalium* (GEKS = 0.130) from the socially complex *M. xanthus* (0.448) by more than a factor of three. The two species with rudimentary multicellular features (*Anabaena*, *M. xanthus*) are distinguished from the strictly unicellular organisms primarily by their *K* and *E* components. This demonstrates that GEKS resolves gradations of complexity even within prokaryotes, a group often treated as uniformly “simple.”

C.2 Model 2: The Vertebrate Radiation

This model traces the classical vertebrate evolutionary sequence—fish, amphibian, reptile, bird, mammal—to test whether GEKS increases monotonically along a well-studied phylogenetic trajectory.

Organisms. Zebrafish (*Danio rerio*), African clawed frog (*Xenopus laevis*), green anole (*Anolis carolinensis*), chicken (*Gallus gallus*), mouse (*Mus musculus*).

Component estimates. *S*. All vertebrates have large genomes with broadly similar nucleotide-level entropy. Discrimination comes from higher-level features: number of cell types (~120 in fish, ~150 in amphibians, ~170 in reptiles, ~200 in birds, ~210 in mammals), protein family diversity, and alternative splicing complexity.

G. Increases with regulatory complexity: mammals have the most elaborate gene regulatory networks, enhancer landscapes, and epigenetic regulation.

E. Increases along the vertebrate lineage as innovations accumulate: jaws (gnathostomes), lungs (tetrapods), amniotic egg (amniotes), feathers and flight (birds), endothermy

and placentation (mammals).

K . Fish through reptiles are Level 5–6 (Organ systems to Sensory integration); chicken is Level 5.5 (some avian species show complex cognition, but the domestic chicken is primarily at the organ-system level with basic sensory integration); mouse is Level 6.5–7 (social learning, problem-solving, approaching cognitive modelling).

Table 6: Toy Model 2: Vertebrate radiation (balanced weights).

Organism	S	G	E	K	GEKS
Zebrafish	0.60	0.55	0.48	0.60	0.558
Frog	0.63	0.58	0.55	0.60	0.590
Anole	0.65	0.62	0.62	0.60	0.623
Chicken	0.68	0.67	0.70	0.66	0.678
Mouse	0.75	0.78	0.80	0.70	0.758

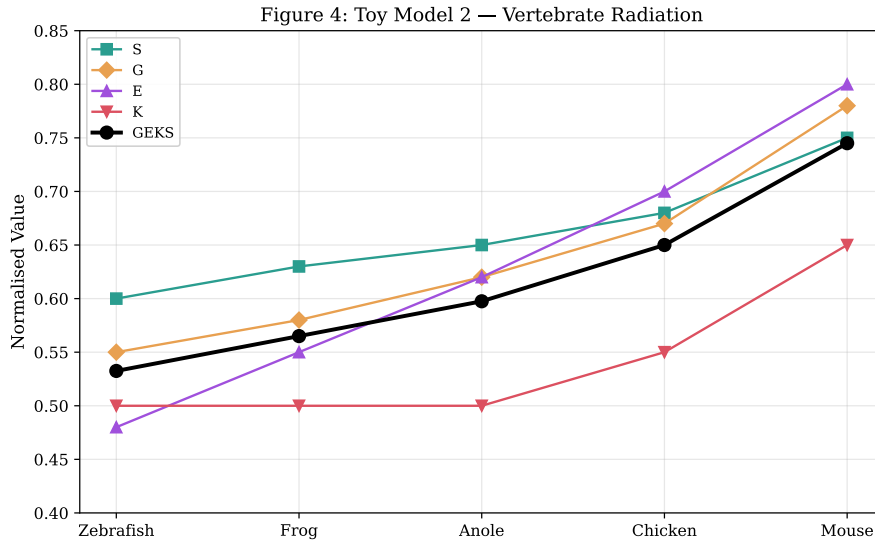


Figure 4: Toy Model 2: GEKS components across the vertebrate radiation. All four components and the composite score (black) increase monotonically, with the steepest gradient in E (evolutionary innovation).

Key observations. GEKS increases monotonically along the vertebrate radiation, consistent with the prediction that more phylogenetically derived lineages accumulate higher composite complexity. The increase is driven primarily by E (cumulative innovations) and G (regulatory depth), with K contributing a smaller gradient. The chicken scores higher than the anole largely due to E (feathers, endothermy, flight-related skeletal modifications) rather than K (both are Level 5–6 organisms). This model validates the falsifiability claim: the predicted monotonic increase is empirically testable and could, in principle, be refuted by data.

C.3 Model 3: The Cognitive Lineup

This model focuses on organisms selected for their cognitive abilities, specifically to demonstrate the discriminatory power of the Kriger component. The organisms span a wide range of phylogenetic positions but are united by the possession of notable cognitive capacities.

Organisms. Ant (*Atta cephalotes*, leaf-cutter), octopus (*Octopus vulgaris*), crow (*Corvus moneduloides*, New Caledonian), chimpanzee (*Pan troglodytes*), human (*Homo sapiens*).

Table 7: Toy Model 3: Cognitive lineup (balanced weights).

Organism	S	G	E	K	GEKS	K level
Ant (individual)	0.65	0.62	0.68	0.65	0.650	6.5
Octopus	0.70	0.68	0.62	0.70	0.675	7
Crow	0.76	0.72	0.74	0.70	0.730	7
Chimpanzee	0.82	0.84	0.86	0.80	0.830	8
Human	0.82	0.85	0.88	1.00	0.888	10

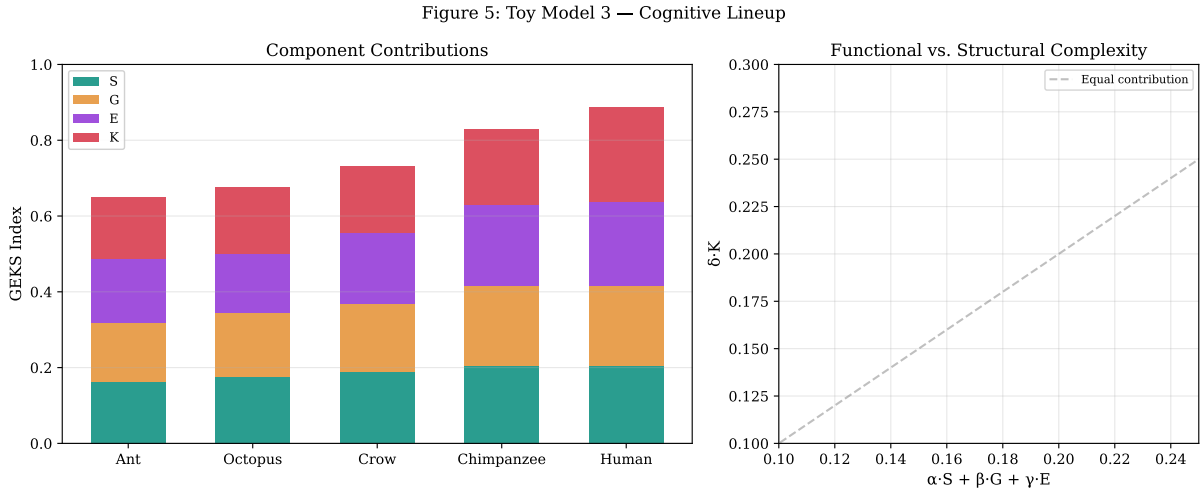


Figure 5: Toy Model 3: Cognitive lineup. Left: stacked component contributions. Right: functional ($\delta \cdot K$) vs. structural ($\alpha \cdot S + \beta \cdot G + \gamma \cdot E$) contribution, showing the partial independence of functional and structural complexity.

Key observations. The ant and octopus receive similar GEKS scores (0.650 vs. 0.675) and comparable K levels (6.5 and 7), despite being separated by over 600 million years of independent evolution. They arrive at similar functional complexity through entirely different routes: the ant through social organisation (an arthropod with colonial agriculture), the octopus through individual neural sophistication (a mollusc with a distributed

nervous system and problem-solving abilities). The crow scores $K = 0.70$ (Level 7, Cognitive modelling), while the chimpanzee reaches $K = 0.80$ (Level 8, Social cognition); the chimpanzee achieves a higher overall GEKS (0.830 vs. 0.730) due to superior scores on all components. Only the human reaches $K = 1.00$ (Level 10, Cultural reflexivity), and this single component accounts for the largest single-step increase in the table (from 0.830 to 0.888).

This model demonstrates that the Kriger component captures a dimension of complexity that is partially independent of the other three: high cognitive function does not require high morphological innovation (the crow is anatomically a “standard bird”), and high morphological complexity does not guarantee high cognition (many morphologically complex organisms are behaviourally simple).

C.4 Model 4: Convergent Complexity

This model selects three organisms that achieve similar GEKS scores through radically different evolutionary strategies, demonstrating the multidimensionality of the index and its capacity to distinguish qualitatively different “kinds” of complexity.

Organisms. New Caledonian crow (*Corvus moneduloides*), greater horseshoe bat (*Rhinolophus ferrumequinum*), bottlenose dolphin (*Tursiops truncatus*).

Table 8: Toy Model 4: Convergent complexity—similar scores, different profiles (balanced weights).

Organism	S	G	E	K	GEKS
Crow	0.76	0.72	0.74	0.70	0.730
Bat	0.80	0.78	0.85	0.50	0.733
Dolphin	0.80	0.80	0.72	0.80	0.780

Figure 2: Convergent Complexity — Same Score, Different Profiles

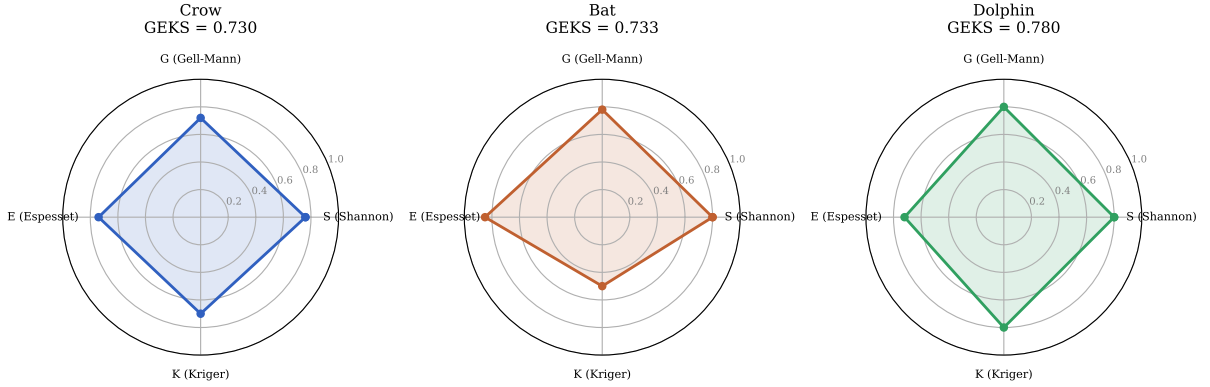


Figure 6: Toy Model 4: Radar plots for three organisms with nearly identical GEKS scores but qualitatively different complexity profiles. The crow excels on K (cognitive), the bat on E (morphological innovation), and the dolphin on $S + G$ (informational/structural richness).

Analysis of profiles. All three organisms fall within a narrow GEKS band (0.730–0.780), yet their complexity profiles are strikingly different:

The crow achieves its score through cognitive sophistication ($K = 0.70$, Level 7: tool manufacture, causal reasoning, future planning) despite being a relatively “standard” vertebrate morphologically. Its E is moderate—birds have accumulated many innovations (feathers, flight, endothermy), but fewer than mammals.

The bat achieves a comparable score through an exceptional concentration of morphological innovations ($E = 0.85$: powered flight via modified forelimbs, laryngeal echolocation, hibernation physiology, specialised auditory cortex). However, its functional level is lower ($K = 0.50$, Level 5: organ systems)—bats are not noted for tool use or causal reasoning.

The dolphin takes a third route: high informational and structural complexity ($S = 0.80$, $G = 0.80$) reflecting a large, heavily folded neocortex, combined with high cognitive function ($K = 0.80$, Level 8: social cognition, mirror self-recognition, cooperative hunting, vocal learning). Its E is somewhat lower (0.72) because cetaceans have *lost* many ancestral mammalian innovations (limbs, fur, external ears) in adapting to an aquatic environment.

The complexity triangle. These three profiles can be visualised as occupying distinct positions in GEKS component space:

Organism	Dominant axis	Strategy
Crow	K (functional)	Cognitive sophistication
Bat	E (evolutionary)	Morphological innovation
Dolphin	$S + G$ (informational)	Neural/structural richness

This model demonstrates the core value proposition of a multidimensional index: a single scalar score compresses real differences into apparent similarity. The full GEKS profile (S, G, E, K) reveals that there are multiple, qualitatively distinct “routes” to comparable levels of overall complexity. Convergent GEKS scores do not imply convergent biology; they imply that different evolutionary strategies can achieve comparable levels of total accumulated complexity through different combinations of informational diversity, structural organisation, morphological innovation, and functional sophistication.